31. Stretching and pain — back, neck, arms, legs, feet, cramps and heavy effort

Oracle OS guide. Based on US Army FM 7-22 and ATP 7-22.02 (**illustrated** stretching and recovery drills), NIH/NIA "Exercise & Physical Activity", NHS (pain, cramps, sprains) and IFRC. For bleeding, tourniquets, fractures and wounds see also "First aid" and the illustrated "Stop the Bleed" PDF.

Red flags — get medical help (don't stretch it out)

- Back pain with **numbness around the genitals/buttocks**, loss of bladder or bowel control, or worsening leg weakness → **emergency** (nerve compression).
- Chest, left arm or jaw pain, breathlessness → could be the heart (999/112).
- **Swollen, warm, red, painful calf** → possible blood clot; don't massage.
- Pain after a fall or blow with deformity, can't bear weight, bone visible → fracture: immobilise.
- Pain with fever, unexplained weight loss, night pain that won't ease, or after cancer → medical assessment.

How to stretch well

1. **Warm up** 5 minutes first (walking, arm circles, marching on the spot). Cold muscle tears more easily.
2. Stretch to **comfortable tension**, never sharp pain. No bouncing.
3. **Hold 20–30 seconds**, 2–3 times each side. Breathe slowly, ease deeper on the out-breath.
4. Daily beats occasional. After a big effort, stretch gently and walk.
5. Pain that worsens during or the next day → do less.

Lower back

- **Knee to chest**: lying on your back, pull one knee to the chest, 20–30 s; switch; then both.
- **Pelvic tilt**: lying, knees bent, flatten the low back into the floor by tightening the tummy, 5 s, 10 times.
- **Cat–cow**: on all fours, round the back up (breathe out), then let the belly sink and look ahead (breathe in). 10 slow reps.
- **Child's pose**: kneel, sit back on the heels, arms stretched forward on the floor, forehead down. 30–60 s.
- **Lying rotation**: knees bent together, let them fall to one side, shoulders down, 20 s; other side.
- **Bridge**: lying, knees bent, lift the hips until knees-hips-shoulders line up, 5 s, 10 times (strong glutes protect the back).
- **Prone press-up** (disc-type pain that eases when standing): lie face down, prop on forearms 30 s; then on straight arms, hips down, 10 times. Stop if pain travels down the leg.

- **Figure-4 (piriformis / sciatic)**: lying, right ankle over left knee, pull the left thigh to the chest, 30 s; switch.
- **Hamstrings**: lying, leg straight up with a towel around the foot, 30 s.
- **Hip flexor**: half-kneeling, push the hips forward keeping the trunk upright, 30 s.
- In common low back pain, **staying active** (walking) recovers faster than bed rest. Heat eases stiffness.

Neck and shoulders

- **Chin tucks**: 5 s, 10 times — against "phone head".
- **Side bend**: ear to shoulder, the same-side hand pulls gently, 20 s each side.
- **Rotation**: look over one shoulder, then the other, slowly.
- **Upper trapezius**: hand behind the back, tilt the head the other way and slightly down, 20 s.
- **Shoulder rolls** backwards 10 times; **squeeze the shoulder blades** 5 s, 10 times.
- **Doorway chest stretch**: forearms on the door frame, step forward, 30 s.
- **Cross-body shoulder**: straight arm across the chest, pulled by the other arm, 20–30 s.
- **Pendulum** (sore shoulder): lean on a table and let the arm hang and make small circles, 1 minute.

Arms, elbows, wrists and hands

- **Wrist flexors**: arm straight, palm up, pull the fingers down with the other hand, 20–30 s.
- **Wrist extensors** (tennis elbow): arm straight, palm down, bend the wrist down, 20–30 s.
- **Triceps**: hand behind the head, the other hand pushes the elbow, 20 s.
- **Biceps**: hands clasped behind the back, arms straight, lift gently.
- **Tendon glides**: open hand, hook fist, full fist, tabletop — 10 times (tingling hands, repetitive work).
- **Tennis elbow** (pain on the outside of the elbow): slow strengthening — hold a light weight (bottle), palm down, lower the wrist over 3–4 s, 3×15 daily for 6–12 weeks.
- Thumb massage of the forearm relieves overload from carrying and gripping.

Legs

- **Quadriceps**: standing, supported, pull the heel to the bottom, knees together, 30 s.
- **Calves**: hands on a wall, back leg straight, heel down, 30 s; then with the back knee bent (soleus).
- **Adductors**: seated, soles together, knees down (butterfly), 30 s.
- **Glutes**: seated, cross one leg over the other and hug the knee to the chest, 30 s.

Feet

- **Plantar fasciitis** (heel pain on the first steps in the morning): calf stretches (above) several times a day; **towel stretch** around the sole pulling the toes toward you, 30 s, before getting up; **roll the sole** over a bottle (frozen helps), 2 minutes; supportive footwear, avoid barefoot on hard floors.
- **Toes**: pick up a towel with the toes; spread the toes; ankle circles.
- **Blisters**: don't pop if you can avoid it; protect with a plaster; if it bursts, wash, leave the skin, cover. Prevent: dry socks, well-laced boots, tape over friction spots.
- **Trench foot** (cold wet feet for many hours): dry, warm slowly, elevate, change socks daily.

Cramps

- **At the moment**: stretch the muscle the opposite way and hold until it eases — calf: straighten the knee and pull the toes toward you; front thigh: bend the knee pulling the heel up; sole: pull the toes up. Massage and walk a little.
- **Prevent**: drink through the day; in heat and hard work replace salts too (home ORS: 1 L water + 6 level teaspoons sugar + ½ level teaspoon salt); stretch calves before bed; warm up before effort.
- Frequent cramps, with weakness, swelling or after starting a new medicine → doctor.

After carrying someone or heavy loads for hours

- **Technique**: keep the load close, bend knees and hips (not the spine), back straight, lift with the legs, **don't twist** — turn your feet. Share the weight (2- or 4-person stretcher), switch sides every few minutes, rest every 10–15 minutes.
- **Moving casualties**: a clothing or blanket drag costs less than carrying in your arms; improvised stretcher (see "Search and rescue").
- **Recover**: walk slowly 5–10 min, stretch back, shoulders, forearms, quads and calves; drink and eat (protein + carbohydrate); sleep.
- **Delayed muscle soreness** (24–72 h later, stiff sore muscles): normal; gentle movement, heat, massage; no need for complete rest.
- **Muscle strain** (sudden pain during effort, sometimes bruising): first 2–3 days **protect** (offload), **elevate**, **compress** with a bandage, wrapped cold 15–20 min; then **move again** gradually and progressively without strong pain.

Bone and joint pain

- Pain at one spot on a bone, worse on weight-bearing after long marches → possible **stress fracture** (foot, shin): rest from impact 4–8 weeks.
- Sprain (twisted ankle): protect, elevate, compress, cold; weight-bear as tolerated; splint or figure-of-8 bandage.
- Arthritic knee or hip: regular gentle movement (walking, cycling, water) helps more than stopping.

Open wounds (summary)

- Firm direct pressure to stop bleeding (10 min). Life-threatening limb bleeding that won't stop: **tourniquet 5–7 cm above, tighten until it stops, write the time** (see "First aid" and "Stop the Bleed").
- Irrigate with plenty of clean water, remove dirt, cover with a clean dressing; change daily. Infection (spreading redness, heat, pus, fever) → antibiotics/doctor. Keep tetanus current.

Pain relief with and without medicines

- Heat (hot-water bottle, warm cloth) for stiffness and spasm; wrapped cold 15–20 min for a recent injury and swelling.
- Paracetamol (adult: 500 mg–1 g, max 4 g/day; less with liver disease or alcohol). Ibuprofen (adult: 200–400 mg with food, max 1.2 g/day without prescription) — **avoid** with ulcers, kidney disease, late pregnancy, blood thinners or asthma worsened by anti-inflammatories. Always check the leaflet.